Understanding the Nature of Environmental Science

Lesson 1

Understanding the Nature of Environmental Science

1. Environmental Principles
2. Environmental Ethics
3. Ecological Concept

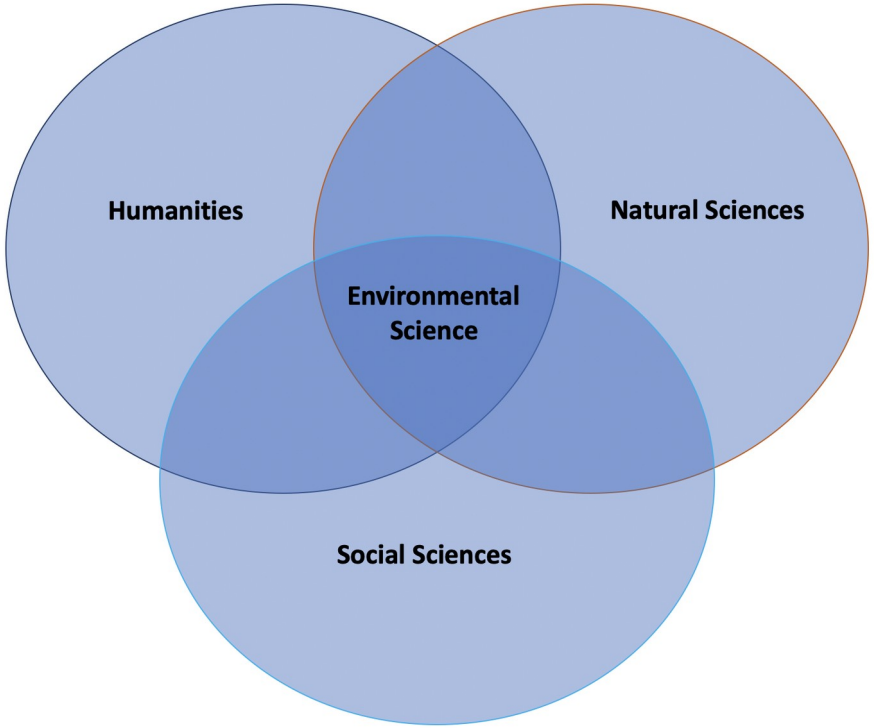
Learning Objectives

At the end of this lesson, students should be able to:

1. Define environmental science and explain its importance.
2. Identify key environmental principles.
3. Understand environmental ethics and their role in decision-making.
4. Apply concepts to real-life environmental issues.
5. Identify key ecological concepts.

Environmental Science

- **Environmental science** is the dynamic, **interdisciplinary** study of the interaction of living and non-living parts of the environment, with special focus on the impact of **humans on the environment.**



Environmental science includes disciplines in the physical sciences (like geology, soil science, physical geography, chemistry, and atmospheric sciences), life sciences (like ecology, conservation biology, restoration biology, and population biology), social sciences (like human geography, economics, law, political science, and anthropology), and humanities (philosophy, ethics).

Environmental Principles

To understand how the world works, we follow several core ecological principles. These serve as the "ground rules" for nature.

Environmental Principles

1. Everything is connected to everything else (Interdependence).
2. All forms of life are important (Diversity).
3. Nature knows best.
4. Ours is a finite earth (Resource Management).
5. Nature is beautiful, and we are stewards of God’s creation.
6. Everything changes.
7. Everything must go somewhere

Environmental Principles

1. Everything is connected to everything else (Interdependence)

- Nature is a complex web. If you change one part of an ecosystem (e.g., removing a predator), it creates a ripple effect that impacts all other parts (e.g., overpopulation of prey and depletion of vegetation).

2. All forms of life are important (Diversity)

- Biodiversity is the backbone of ecosystem resilience. A variety of species ensures that an ecosystem can survive diseases, climate shifts, or disasters.

Environmental Principles

3. Nature knows best

- Nature has evolved self-regulating mechanisms over billions of years. When humans attempt to "improve" nature, it often leads to unforeseen negative consequences.

4. Ours is a finite earth (Resource Management)

- Earth’s resources, water, minerals, and space, are not infinite. We must live within the "carrying capacity" of our planet.

Environmental Principles

5. Nature is beautiful and we are stewards of God's creation

- This principle bridges science and morality. It suggests that humans have a responsibility (Stewardship) to protect the environment not just for its utility, but for its intrinsic value.

6. Everything changes

- Nature is dynamic, not static. Ecosystems go through succession, and species evolve. Human-induced change, however, often happens too fast for nature to adapt.

7. Everything must go somewhere

- Waste does not disappear; it accumulates in land, air, or water.

How does throwing plastic affect the environment?



Environmental Ethics

What is the **right** thing to do for nature?

Environmental Ethics

Environmental ethics is the branch of philosophy that studies the **moral relationship** of **human beings to the environment**.

It is the study of **how humans should treat the environment**.

Three Primary Worldviews

1. Anthropocentrism (Human-Centered)
2. Biocentrism (Life-Centered)
3. Ecocentrism (System-Centered)

1. Anthropocentrism (Human-Centered)

Humans are the most significant entity in the universe.

Nature is used for human benefit.

- **Perspective:** Nature has value only in how it serves human needs (economic, recreational, or medical).

Example

- Protecting a forest only because it provides timber or clean water for a city.
- Cutting forests for development.

2. Biocentrism (Life-Centered)

All living things have intrinsic value, regardless of their usefulness to humans.

Humans are **not** superior.

- **Perspective:** Every individual organism, from a whale to a blade of grass, has a "right to exist.“

Example

- Opposing a dam because it would destroy a unique species of fish, even if the dam provides electricity.

3. Ecocentrism (Nature-Centered)

The entire ecosystem (including non-living components like soil and water) is the focus of moral concern.

Entire ecosystems are valuable.

Focus on balance and sustainability.

- **Perspective:** It is not just about individuals, but about the integrity of the whole system.

- **Example:**
  - A community stops building resorts near a coral reef to **preserve marine life and ecosystem health**.
  - A government decides to **protect a forest from logging**, even if it could generate profit, because the forest is important for wildlife habitats, clean air and water, and ecological balance.

Core Ethical Principles

- **Stewardship** – Humans must care for the Earth.
- **Sustainability** – Use resources without harming future generations.
- **Responsibility** – Be accountable for environmental actions.
- **Justice** – Fair sharing of resources.

Preservation versus Conservation

Conservation (The Wise-Use Approach)

- Championed by **Gifford Pinchot**.
- It argues that we should use resources sustainably "**for the greatest good for the greatest number for the longest time.“**

Preservation (The Hands-Off Approach)

- Championed by **John Muir.**
- It argues that nature should be protected in its pristine state, untouched by human development.

Questions for Reflection

1. Which of the three worldviews (Anthropocentrism, Biocentrism, or Ecocentrism) best describes your own view?

Case Study

- A company plans to build a factory near a river.

1. Is this ethical?
2. What environmental impacts may occur?
3. Which ethical view supports or opposes it?



Importance of Environmental Science

Environmental science helps:

1. Protect natural resources.
2. Prevent environmental damage.
3. Promote sustainable living.
4. Improve human health.



Ecological Concepts

What is Ecology?

- Ecology is the **study of interactions** between organisms and their environment.

Levels of Organization in Ecology

From smallest to largest:

- **Organism** – a **single** living individual.
- **Population** – a group of the **same** species living in one area.
- **Community** – **different** species living together
- **Ecosystem** – living (biotic) + non-living (abiotic) components
- **Biome** – large area with similar climate (e.g., desert, forest)
- **Biosphere** – all ecosystems on Earth

Biotic and Abiotic Factors

- **Biotic factors** – living components (plants, animals, bacteria)
- **Abiotic factors** – non-living components (sunlight, water, temperature, soil)

- Organisms depend on:
  - Temperature
  - Water
  - Light
  - Soil nutrients

If these change, organisms may:

- Adapt
- Migrate
- Become extinct

Habitat versus Niche

- **Habitat** – where an **organism lives.**
- **Niche** – the **role or function** of an organism in its environment.

Example:

- A frog’s habitat = pond
- Its niche = predator of insects and prey for snakes



Bird Niches.

Each of these species of birds has a beak that suits it for its niche.

- The long, slender beak of the nectarivore allows it to sip liquid nectar from flowers.
- The short, sturdy beak of the granivore allows it to crush hard, tough grains.

Next Meeting

Advance Study Task



Before our next meeting, review and familiarize yourself with the organisms that I will provide.



You do **NOT** need to write anything, just be ready to recognize and understand them.



Be prepared to use this knowledge in our next activity.

Organisms

- Wheat
- Deer
- Fly maggot
- Pine tree
- Algae
- Bear
- Hawk
- Rabbit
- Millipede
- Bacteria
- Sparrow
- Mouse
- Frog
- Snake

Organisms

- Bacteria & Human intestines
- Spanish moss & Trees
- Dogs & Fleas
- Squirrels & Trees
- Whales & Barnacles
- Butterfly & Flower
- Figs & Fig wasps
- Remora & Sharks
- Tawny owl & Mice
- Field digger wasp & Fly
- Clown fish & Sea anemone
- Venus fly trap & Insects
- Rafflesia plant & Vine
- Ants & Acacia trees
- Tapeworm & Mammal
- Trees & Orchids
- Human head & Lice
- Trees & Mistletoe
- Yucca moth & Yucca plant
- Eagle & Rabbit